



Machine Learning vs. Deep Learning

A Strategic Methodology Evaluation Portfolio

Two real-world case studies — automated home valuation and autonomous vehicle perception — showing why the right algorithm depends entirely on the shape of the data.

Two Problems, Two Very Different Toolkits



Case Study 1

Automated Real Estate Valuation

- Example: Zillow's Zestimate
- Approach: Multiple Linear Regression & Gradient Boosted Decision Trees
- Data: clean, structured, tabular

WINNER: TRADITIONAL ML



Case Study 2

Autonomous Vehicle Perception

- Examples: Tesla Autopilot, Waymo
- Approach: Deep Convolutional Neural Networks (CNNs)
- Data: massive, unstructured video streams

WINNER: DEEP LEARNING



CASE STUDY 1

Automated Real Estate Valuation

Zillow's Zestimate prices millions of U.S. homes automatically — no appraiser visit required.

The Raw Material: Structured, Tabular Data

Every home is a tidy row of human-defined features. This data shape decides the algorithm.



Square footage

A precise, numeric physical measurement



Bedrooms & bathrooms

Simple counts, directly comparable across homes



Location (ZIP code)

Categorical signal that anchors market value



Year built

A clean proxy for condition and style

Algorithms of choice: Multiple Linear Regression · Gradient Boosted Decision Trees

Why Traditional ML Is the Right Tool



Built for tabular data

Regression and boosted trees thrive when features are explicitly defined by humans — square footage in, price out.



Computationally lean

Feature weights are calculated efficiently with modest hardware. No GPU farm, no massive training bill.



Highly explainable

Analysts and clients can see exactly why a home was valued at a specific price, feature by feature.

The result: fast, auditable valuations that a business can defend to any customer or regulator.

Why Deep Learning Would Be Overkill



Mathematically inefficient

Deep neural nets are designed to mine massive unstructured data like raw video or audio. Pointing one at a small tabular dataset wastes enormous compute for no accuracy gain.



Prone to overfitting

With a structured, relatively small dataset, a deep network tends to memorize the training data instead of learning generalizable pricing trends — great on paper, wrong in the market.



A black box

It becomes incredibly difficult to explain to a homeowner or an auditor precisely how the network arrived at a specific property value.

Verdict: for housing prices, deep learning adds cost and opacity — not intelligence.

The Raw Material: Unstructured Visual Chaos

No tidy rows here — just torrents of raw pixels that must become split-second decisions.



Multiple live camera feeds

High-dimensional video streamed simultaneously from around the vehicle



Pedestrians & hazards

Detected across infinite variations of lighting, weather, angle, and clothing



Lane boundaries & signs

Geometry and context read directly from raw pixel arrays



Real-time stakes

Perceptual inferences must arrive in milliseconds — lives depend on it

Algorithm of choice: Deep Convolutional Neural Networks (CNNs)



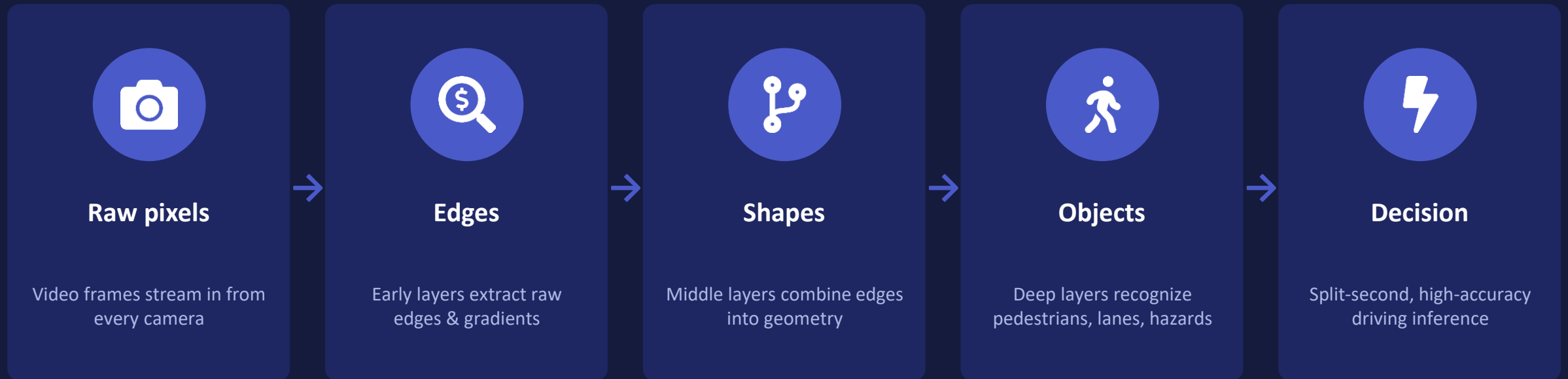
CASE STUDY 2

Autonomous Vehicle Perception

Systems like Tesla Autopilot and Waymo must see and understand the road in real time.

Why Deep Learning Is Uniquely Suited

A CNN's hidden layers learn their own features — no human hand-coding required. Each layer builds on the last:



- No manual feature engineering — the network teaches itself what matters
- Scales exceptionally well as visual data volume grows into the billions of frames
- Delivers the rapid, high-accuracy perception real-time driving demands

Why Traditional ML Breaks Down on the Road



Can't ingest raw pixels

Classical algorithms cannot natively process unstructured pixel arrays at this scale — they expect neat, pre-defined feature columns.



Impossible feature engineering

A human would have to hand-code every possible feature of a pedestrian across all lighting, weather, angles, and clothing — the real world's variance is infinite.



No layered architecture

Tabular algorithms lack the deep architectural complexity to interpret high-dimensional video streams in real time.



Unsafe by design

The combined result: a perception system that would be completely unsafe for autonomous navigation.

Verdict: manual feature extraction cannot keep up with an infinitely variable world.

Head-to-Head: Choosing the Right Method

	Traditional ML	Deep Learning
Data type	Structured, tabular rows	Unstructured pixels, video, audio
Feature engineering	Defined by human experts	Learned automatically by hidden layers
Compute needs	Modest — efficient math	Massive — scales with data volume
Explainability	High — clear feature weights	Low — operates as a black box
Dataset size	Small–medium works well	Needs vast data to avoid overfitting
Best-fit example	Home price prediction	Autonomous vehicle perception

Match the Method to the Data



Choose Traditional ML when...

- Data is structured and tabular
- Features are human-defined and clear
- Explainability and audits matter
- Compute budgets are modest



Choose Deep Learning when...

- Data is unstructured — video, images, audio
- Manual feature engineering is impossible
- Data volume is massive and growing
- Real-time perception accuracy is critical

There is no universally superior algorithm — only the right tool for the shape, scale, and stakes of the problem.